Review of Groundwater Potential Storage and Recharge Zone Map Delineation Using Statistics based Hydrological and Machine Learning based Artificial Intelligent Models

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***Abstract*— Groundwater accounts for 63% of agriculture irrigation and 80% of household water supplies. In Many parts of the country, the water table is going down by 1 to 2 meters per year due to over utilization. It may result in up to a 20 percent decrease in food production. Given the huge impact this invisible resource has on the economy, environment, and society, we need to improve the scientific understanding, estimation, and governance of groundwater. The creation of groundwater's possible storage area (GWPSZ) and regional recharge (GWRZ) zone maps can be helpful in this regard. They depend on statistics-based, machine learning (ML) based, and hybrid models. This paper reviews the work done by multiple researchers who have used geospatial techniques using satellite imagery sensing analytics in GIS, followed by AHP or Multi - Influencing Factors (MIF) pairwise comparison to characterize, forecast GW levels, and generate GWPSZ and recharge GWRZ maps. We also reviewed the research on historical aquifer data using ML-based regression analysis, random forest (RF), supervised algorithms like support vector machine, non-parametric ML algorithm decision tree model, and ensemble hybrid multi-wavelet ANN models for the prediction of the GWL variability and storage loss/deceleration. Though some papers focused on the use cases like irrigation scheduling and predicting geothermal well locations or designing community cooling hubs, a comprehensive approach for village-level community water demand and supply assessment and decision-making is missing.**

Keywords*— *Groundwater estimation, groundwater potential storage zone, recharge zone, hydrological model, machine learning

I. INTRODUCTION

"Water water, everywhere, But Nor any drop to drink" are the famous lines from the poem by Samuel Taylor [1]. This is true because only approximately 2.78% of the aggregated volume of water, from all the sources, on the earth is freshwater [2]. This includes 0.6% of groundwater available beneath the earth's surface. The calculated quantity of groundwater is approximately 30 times the total freshwater volume in lakes and reservoirs [3]. But we can economically extract only the water to 800 m depth using the present drilling technology [4]. Around 40% of the irrigated area across the globe is supplied by groundwater (GW), enabling 13% of total global food production [5 – 6]. Around 2 billion people globally consume underground water for all their basic needs for daily living [7]. In India, groundwater accounts for 63% of agriculture irrigation and 80% of rural and urban household water supplies [8].

Water shortage has always been a constant challenge in drought sensitive geographies This impacts the food production and home usage. All combined sources of water is not enough to meet village level needs. As rain is becoming unpredictable across the globe, it affects the rainfed agriculture. Crop production is becoming unsustainable and reliance on the repeat sowing, nutrient management and improved pest/disease management is increasing, causing food insecurity concerns [9]. Besides these, various water courses like streams, rivulet, tributaries are disappearing or drying up due to natural phenomenon or human led activities. This is leading to increase the reliance on the groundwater (GW) [10]. Saltwater intrusion is another risk in the coastal aquifer [11]. Groundwater acts as a risk mitigation tool towards drought and changing climate [12 – 13]. Therefore, there is thrust on new groundwater aquifers discovery and their potential storage estimation to meet the newer demands for the feasible water usage for household, agriculture factories, and manufacturing sector developments, effective groundwater governance, and identifying appropriate aquifers and their recharge strategies [14]. Satellite imagery-based remote sensing technique helps in generating proxy data to identify the aquifer zones, especially under data-scarce conditions [15]. This site-specific information can also be generated from the geographic information system (GIS) environment various geospatial thematic layer attributes [16]. Remote sensing techniques yield vast insights into hydrologic attributes [17].

Water Use Group (WUG) should be aware of the sustainable water table level and monitor their usage does not exceed replenishable level which is a function of the multi-year average recharge. [18-19].

The groundwater potential (GWP) maps help in the above decision-making. Artificial Intelligence and Machine Learning (AIML), remote sensing using satellite-based imagery-GIS are popular techniques in estimating groundwater storage and recharge potentials and delineating GWP zone maps [20]. Delineation of these maps in groundwater potential storage (GWPSZ) and recharge (GWRZ) zone maps and assigning them appropriate weights can help create a region-specific groundwater abstraction and recharge structure creation strategy [21].

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This research is useful in developing data driven feasible groundwater management tactics for administrators for geographical placement of new wells, divulge the impact of human action and pursuits on the groundwater resources. This enabled generating the over-exploitation risk and the development and design policy and framework of zonal water resources administration services [22 – 24].

The present paper summarizes our literature research on groundwater potential estimation. Section two covers the important hydrological parameters associated with groundwater estimation. Section three covers the major techniques of estimation of the groundwater storage and recharge potential. Section four provides the conclusion and future research areas.

II. Topographical and Geological parameters influencing Groundwater Potential

The region from topsoil to groundwater is divided into four layers i.e., the unsaturated zone (UZ), capillary fringe (CF), groundwater table (GWT), and saturated zone (SZ) [25]. UZ is just below the topsoil surface, and it contains a mix of water and air in the pores or spaces. SZ is largely formed of rocks and all the rock fractures or pores are saturated with water. CF is the intermediate zone between UZ, and SZ, and the lower portion of the CF are known as the GWT. Hydraulic attributes of the size and shape of these void spaces determine the groundwater movement pattern [26]. Aquifers are saturated groundwater zones with high permeability and porosity due to geological formation of sand/limestone, gravel, etc. Unsaturated zone water is consumed by the plants through the process of transpiration, or it may evaporate due to evaporation [27]. The downward displacement of the rest of the water reaches the water table to recharge it. Likewise, due to the capillary action, the water in small capillary duct like formation moves upward towards the surface and is available to the rootzone or appears in the form of a spring or riverbed or is pumped out [28]. Hence, the storage and recharge of the groundwater in terrain is a complex function of the multiple of these hydrological parameters like lithology, geological structures, porosity, slope, drainage density, type and extent of fractures, permeability, topography, landforms, land use/land cover, etc. [29]. Table 1 presents an exhaustive list of these parameters.

Porosity is computed as the volume of associated void space of the earth's material like rock formation or soil to the total volume. This is the measure of the space available for water to flow [30].

Permeability is the ability of water movement through given earth material such as rock or soil. Porosity and permeability are interrelated [31]. Permeability is directly proportional to hydraulic conductivity and viscosity. It is inversely proportional to gravity acceleration (g) and the fluid (water and it's solute) density [32].

Transmissivity is the horizontal flow or discharge rate of the water in the aquifer and is expressed in liters per day per meter. It depends on two parameters i.e., the aquifer thickness and hydraulic conductivity [33].

The slope is the characteristic of steepness [34]. It regulates the percolation of surface water into the soil [35].

TABLE 1. HYDROLOGY INPUT PARAMETERS

Hydrology Parameter Description	**Code**
Altitude/Elevation	AL
Aquifer resistivity	AR
Slope aspect	AS
Aquifer type	AT
Aquifer thickness	ATh
Curvature	C
Convergence index	CI
Drainage	D
Drainage density	DD
Slope (degree/percentage)	DS
Distance to drainage	DtD
Distance to faults	DtF
Distance to roads	DtR
Distance to water bodies/rivers	DtW
Fault/lineament	F
Flow direction.	FD
Geomorphology	Ge
Groundwater depth	GWD
Groundwater Potential Recharge;	GWPR
Groundwater Potential Storage;	GWPS
Groundwater quality	GWQ
Geology (Lithology)	L
Lineament density/Fault density	LD
Slope length	LS
Land use	LU
Land use and Cover	LULC
Normalized difference vegetation index	NDVI
Net recharge	NR
Plan curvature	PlC
Profile curvature	PrC
Rainfall/Precipitation;	R
Runoff coefficient	RC
River density	RD
Relative permeability map	RP
Relative slope position	RSP
Soil	S
Soil depth	SD
Stream power index	SPI
Surface temperature	ST
Sediment transport index	STI
Topographic position index	TPI
Terrain ruggedness index	TRI
Topographic wetness index	TWI
Water table level	WTL

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Slope length and slope aspect or degree are major attributes of the slope. They control the flow velocity and the amount of time the water gets for vertical infiltration.

If the slope is steeper, then the water infiltration rate will be slower. But the water runoff will be faster. This is not desirable. Soils with specific size and texture such as granular property and higher level of porosity, will slow down the surface water runoff, compared to soils having massive structures and lower porosity values. A steep angle of slope means the water flow speed is high. Water runoff will be high, and it will not have enough time to percolate into the soil. Hence, geographical characteristics are of positive surface runoff, but a negative infiltration rate. In such cases, soil erosion is also high, which impacts the soil length. Hence, these tend to negatively impact the groundwater storage potential [36].

These geomorphological features of porosity, permeability (water penetration or absorption capability), frequency and intensity of precipitations, upper surface flow direction, and slope, control the water seepage into underlying soil and hence determine the groundwater storage and recharge potential. Digital Elevation Models (DEM) help in drawing out the above topographical features from multiple sources like DEM, LiDAR, etc. [37 – 38]. They are 3D graphical representation data in the form of a raster grid of terrain elevation values.

As mentioned in Table 1, other important geological parameters affecting potential storage and recharge are Dtf, L, LD, relative P, soil structure or rock folds, etc. Knowledge of the geological features of terrain helps in estimating the potential of groundwater storage in weathered rocks or fractures, sediment types such as barren consolidated rock, or the sediment porosity [39].

Lithology determines the hydraulic conductivity, and porosity and influences the rate of flow [40]. Lineaments are the surficial expression of the ground fault zones, fractures, or lithological limits [41]. Fig. 1 shows the linkages of the various thematic layers hydrological factors and their influence on groundwater storage and recharge.

III. Determination of Groundwater Potential

There are multiple methods to determine the groundwater potential storage, discussed below.

A. Traditional point field measurement methods

In these methods, hydrogeological parameters are obtained based on the field-testing using borehole pumping techniques or aquifer hydraulic testing [42]. They are time-consuming, sporadic, costly, and not scalable.

B. Hydrological process-based groundwater potential estimation – hydrogeological parameters

Popular hydrological methods are [43],

1. Water balanced baseflow LFRA approach.

2. For the stream flow recession curve analysis

In a back calculation, hydrogeological parameters are calculated based on hydrological processes and streamflow data of the rivers discharged by the groundwater aquifers. The recession index (RI) characterized the catchment flow behavior, which was used to calculate transmissivity, conductivity, and potential water yield.

C. Other techniques of groundwater potential estimation

Vertical electrical sounding is an inexpensive way to generate high-resolution spatial maps of aquifer transmission capability and hydraulic conductivity [44]. Based on these and aquifer layer thickness, resistivity, Dar-Zarrouk parameters i.e. (transverse unit) resistance and (longitudinal) conductance are calculated. Dar-Zarrouk parameters help determine the GWL.

Authors [45] explored the tritium injection method to estimate the annual replenishable groundwater potential for an aquifer in India. This method is based on the piston flow and is appropriate to gauze the downward fluidity of moisture from topsoil to water table level and establish a linear regression association between precipitation and recharge.

D. Statistical Models

The most popular statistical model used was Multi-criteria decision analysis (MCDA). MCDA is a statistical decision-making tool that compares multiple conflicting criteria and ranks them based on weight and scores assigned to each criterion based on their importance. This helps in filtering the available options/criteria [46].

Thematic Layers	Code	Type	Infiltration	Overland Flow	GW Recharge - High	Evapotranspiration	Deep Percolation	Storage Volume	High Runoff	Outcrops	Water Transmission
Aquifer Thickness	AT	GW Storage					Y	Y			
DEM - Elevation	DEM	GWPZ-RPZ									
Drainage density	DD	GWPZ-RPZ			Low				High		
Geology	GG	GWPZ-RPZ									
Geomorphology	GMG	GW Storage	Y	Y	Y	Y					
Land Use/Land Cover	LULC	GWPZ-RPZ	Y	Y	Y	Y					
Lineament density	LD	GWPZ-RPZ			High					High	High
Lithology	L	GW Storage					Y	Y			
Plan Curvature	CT	GW Storage									
Rainfall	RF	GWPZ-RPZ									
Roughness	RG	GW Storage									
Slope	SP	GWPZ-RPZ		Steep	No Slope						
Soil	S	GW Storage									
Surface Water Bodies	SWB	GW Storage	Y		Y	Y	Y				
Topographic Position Index	TPI	GW Storage									Y
Topographic Wetness Index	TWI	GW Storage	Y								Y
Stream Power Index	SPI	GWPZ-RPZ	Y								
Faults	F	GW Storage					Y	Y			
Rock Fractures	RF	GW Storage					Y	Y			

Fig. 1. Major factors and their influence on groundwater storage and recharge.

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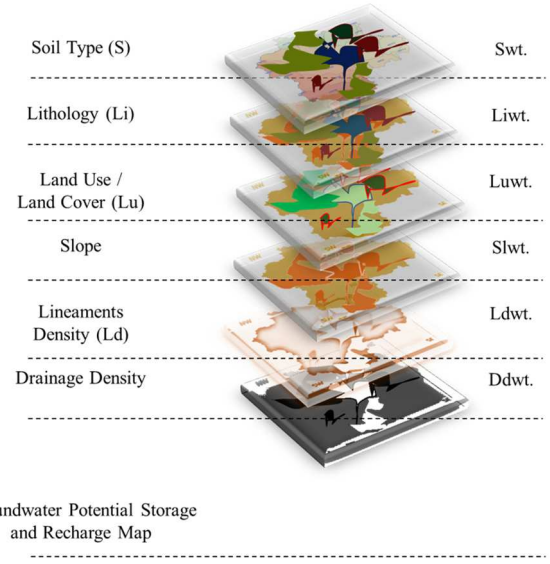


Fig. 2. Thematic layers.

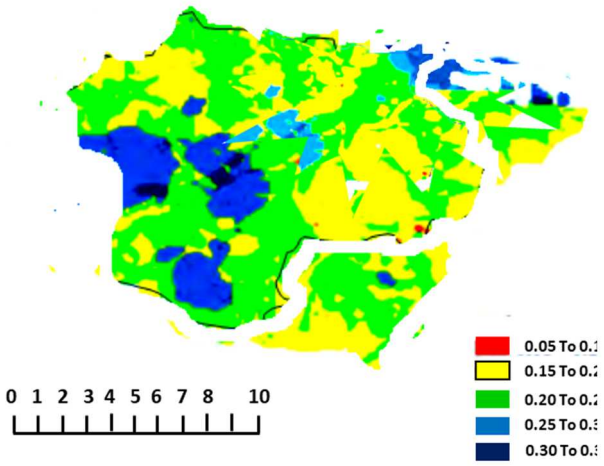


Fig. 3. A sample Groundwater Potential Map.

The Analytical hierarchy process (AHP) is an adaptation of MCDA. In this model, the variables are compared, on a scale of 1 to 9, depending on the effect of each on the other. Hydrologists use MCDM-AHP to filter the hydrology parameters which have maximum influence on the determination of the potential groundwater storage and recharge. This also helps them in picking up the water well spot [47]. This method is faster with better accuracy but has an ounce of subjectivity due to the expert's input to set criteria weight.

Multi-criteria decision analysis (MCDA)-Analytic hierarchy process (AHP), Evidential belief function (EBF), Frequency-ratio (FR), and Weights of evidence (WOE) are the most popular statistical models.

E. Artificial Intelligence and machine-learning (AIML) methodologies

This involves the use of multiple AIML or deep learning approaches to determine the hydrogeological attributes. Widely used models are Boosted regression tree (BRT), Logistic and Linear regression (LR), Random forest (RF), and classifier models like Support vector machine (SVM) are the widely used AIML models in groundwater research.

RF is a supervised ML algorithm using an ensemble of the decision tree as a base to predict the occurrence of a feature or event [48].

Authors have used RF in combination with other ML models such as the Genetic Algorithm (GA) Model [49] or maximum entropy models [50] to estimate the groundwater storage and recharge potential. As a pre-requisite, they use 10 to 14 hydrological variables and data such as geology, slope, land use/land cover, lithology factors, etc. for the study area. Approximately 70% data is used for model training, and 30% data is used for model testing and evaluation. It may generate groundwater potential maps. The model results are performance are assessed against parameters like Root Mean Square Error (RMSE) or receiver operating characteristics (ROC) curve or Area under the curve (AUC). The model results list the input parameters of significance that have more influence on the model outcome. It could be the altitude, slope angle, drainage density, TWI, profile curvature, plan curvature, elevation .etc.

Logistic regression (LR) is another popular AIML model in water research such as ground subsidence [51] and groundwater springs [52]. LR model works on the principle of coefficient correlation calculation between two variables that is dependent and independent variables. Based on this coefficient correlation it predicts the presence (value = 1) or absence (value = 0) of the groundwater. But this model has a risk of getting overfitted.

F. Delineation of groundwater storage using Remote Sensing and GIS

Due to availability of the satellite images and their potential to detect earth-based features, it has become a popular method for groundwater mapping.

This approach involves various thematic layers in the GIS, which include soil, slope characteristics, LULC, faults, geomorphology, lithology, etc as they influence the soil water percolation, residence time, fluidity, and transmissivity attributes of the aquifer (See Fig. 2). Data were collected from various space center programs and satellites, Digital Elevation Model, electricity resistivity tests, automatic weather stations, etc. [53].

MCDA-AHP is used to assign suitable weights to these thematic layers, based on their significance in groundwater storage and prospective recharge. Sensitivity analysis can be done to validate expert opinion in AHP. These were then converted into layer-wise five classes ranging from Very poor, to Moderate, to Very Good. This helps in classifying the study area into multiple zones based on these classifications (See fig. 3). Overlaying of the maps shows how much area of the basin lies in a higher to lower potential zones. This helps central groundwater authorities and planers to think of ways to predict the potential water scarcity and potential corrective actions to be taken to alleviate the situation. Other significant insights such as exploitable and replenishable groundwater can be derived thereafter.

During the processing of the remote sensing data, various AIML techniques such as random forest or support vector machines are useful as data classifiers. ArcGIS is a popular tool for remote sensing data layering and analysis. Major limitations of this approach are high costs of acquisition, storage, and processing of the high-quality, high-resolution satellite images. Cloud cover or tree-top is also a problem

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faced by satellites during the rainy season. Hence, complementary approaches such as the use of ground truth data from borewells, and other hydrological sources will help address some of these challenges.

IV. Conclusion

Due to the change in climate and anthropogenic activities, the dependency on groundwater is increasing. Hence, we need to develop new tools, techniques, and skills to estimate the real-time, region-specific groundwater storage and recharge potentials. This can help water resource planners to plan policies and end users manage withdrawal. An effective estimation approach requires an integrated approach of hydrology, thematic layers, AIML, remote sensing, GIS, and IoT. Hydrological, meteorological and satellite data are key data sources. Popular data analysis and groundwater potential map creation techniques include statistical and ML/DL methods. Widely used machine learning models are RF, LR, BRT, and SVM, or advanced combination of these models ANFIS of differential generation or with genetic algorithm (GA), etc. MCDA with ANP or AHP is the mostly used statistical approach. Other approaches could be FR/EBF/WOE. Use of the remote sensing and GIS is also becoming popular. RMSE, ROC, Kappa index, etc. can be used as the model validation technique.

But the researchers have not been able to find a global unified technique or model due to regional variations such as land use/land cover or runoff, climate change, lack of availability of the input data, and affordable IoT-based sensing technology. This will require the standardization of the research methodologies and domain, the creation of an integrated database of all the observations to date, and future satellite missions with high-resolution data with more frequency and at a lower cost of data access and processing.

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